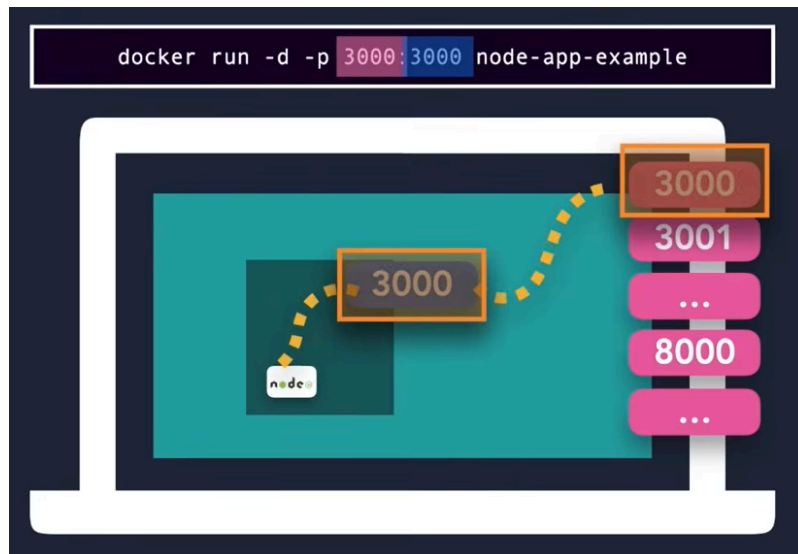


Docker

- **Container** is an instance(object) of an **Image**. like OOP
- Containers in docker are **ephemeral** : they do not have a data persistency (not permanent) like RAM; Because data is not saved in a storage.

<code>docker pull <img-name></code>	pull an image from the hub
<code>docker push <img-name></code>	push an image to the hub
<code>docker run [] <img-name></code> <u>Attributes</u> : • <code>--name <con-name></code> • <code>-d</code> • <code>-p 8080:80</code>	• creating a container to run an image, if does not exist locally it pulls the images from the hub then run it. • <code>--name</code> give it a name • <code>-d</code> for dashboard; to run the image in the background • <code>-p</code> for port; the port to run in browser
<code>docker images</code>	display local running images
<code>docker container ls</code>	• display running containers
<code>docker ps -a</code>	• display all containers
<code>docker exec -it <con-id name> bash</code>	• running an existing container • <code>-it</code> for interactive; it displaying an output of image's terminal if using <u>like ubuntu</u> • <code>bash</code> to open the image's terminal if has <u>like ubuntu</u>
<code>docker rm <con-id name></code>	remove a container
<code>docker rmi <img-name></code>	remove an image
<code>docker start stop <con-id name></code>	start OR stop a container

Port Mapping



Volume

- Is a data storage linking the OS to the image; it saves the data permanently in a place inside the image like DATABASE.
- We can get back the data from the image even if container is deleted and created another one.
- use `\` in code to write on command in multiple lines.

```
docker run -d \  
--name <con-name> \  
-v <vol-name>: /var/lib/<img-  
name>/data \  
<img-name>
```

1. create the container con-name
2. create a volume called vol-name
3. this volume is mapped to this path inside the image
4. the data is exactly saved in this path in the image; so in each DB-image you should know the folder of data to be saved
5. then you can remove the container and get back the data in a another one using the same command.